Experiment 03 — Citation-graph structure

arxivkg · preprocessing · phase 3/4 (graph)

What we did

Characterize the intent-typed citation graph — the substrate phase-6 retrieves over — read from the single Postgres+pgvector store: scale, in/out-degree distributions, per-edge intent/polarity typing, citation growth, and the top-cited papers. Pure SQL + figures, no LLM; heavy aggregations cached to `cache/summary.json`.

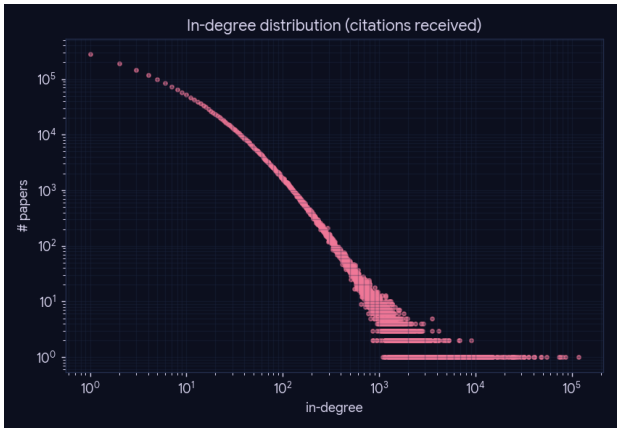
Global counts

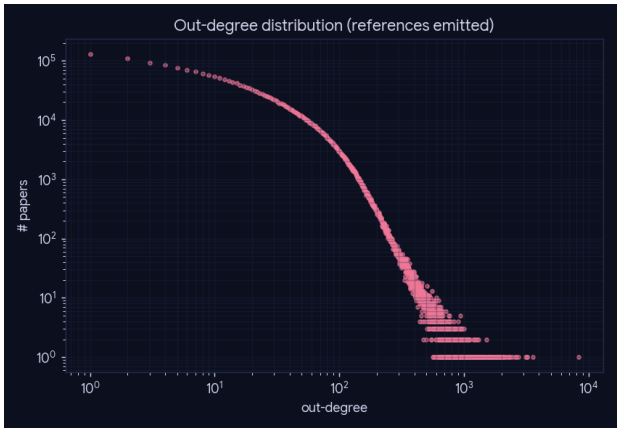
metric	value
papers (nodes)	3,073,288
citation occurrences	176,934,085
resolved occurrences	70,634,716
distinct edges	36,688,723

3,073,288 papers, 176,934,085 citation occurrences, 70,634,716 resolved (39.9%), 36,688,723 distinct edges.

Degree distributions

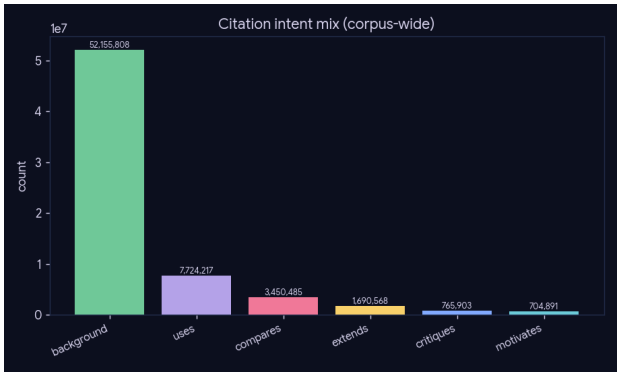
Both in- and out-degree are heavy-tailed (near scale-free): a few papers accrue enormous in-degree while most are cited rarely — the hub structure phase-6's pool induction avoids by ranking *within* an on-topic pool rather than expanding from hubs.

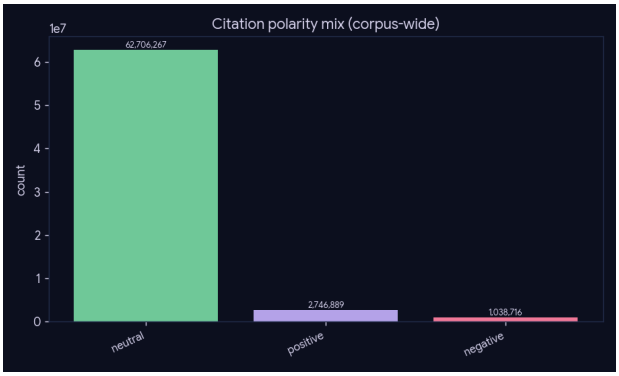




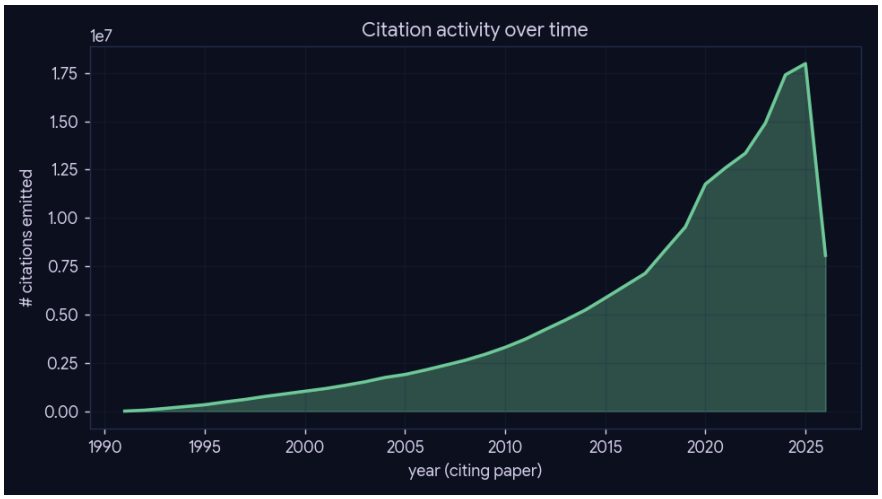
Edge typing (the differentiator)

Every resolved edge carries an intent + polarity. The mix is dominated by `background`/`uses`; the rare relational intents feed phase-6's tensions surfacing.





Citation activity over time



Conclusion

A **36.7M-distinct-edge**, intent-typed, near-scale-free citation graph over **3.07M** papers, whose in-degree ranking recovers the field's landmark papers (ResNet, BERT, Transformer, CLIP, . . . — full table in `results/metrics.md`). The heavy-tailed hubs motivate the "induce within a topical pool" design, and per-edge typing is what distinguishes this substrate from a plain citation graph. In-degree is within-corpus (arXiv→arXiv resolved only).